

Churn Model Training Report

Trained at : 2026-08-19
Training dates : 2026-07-17, 2026-07-22
Holdout date : 2026-07-27
Features : 64 training (9 leakage, 5 dead excluded)
Training samples : 7996 (406 positive)
Holdout samples : 1000 (60 positive)
Class imbalance : 5.08% positive
scale_pos_weight : 18.6946

Metrics (raw probabilities)

Metric	Value
AUC	0.4737
5-fold OOF CV AUC	0.4623
Brier	0.2125
LogLoss (test)	0.614
LogLoss (train)	0.5819
Accuracy (optimal)	0.9
ECE (raw)	0.3437
Optimal threshold	0.5585
F1-optimal threshold	0.3679

Classification & Calibration

Confusion Matrix (optimal threshold)

	Pred 0	Pred 1
Actual 0	894	46
Actual 1	54	6

Precision=0.1154 Recall=0.1 F1=0.1071

Probability Calibration (out-of-fold)

Item	Value
Method	isotonic
ECE raw	0.3799
ECE Platt	0.0101
ECE isotonic	0.0095
ECE after	0.0095
Brier after	0.0565
LogLoss after	0.2279

Feature Importances & Hyperparameters

Top-10 Feature Importances

Feature	Importance
distinct_channels_90d	0.0347
risk_high_value_txn_ratio_90d	0.0314
temp_payday_activity_ratio_90d	0.0306
fin_median_txn_amount_90d	0.0299
behav_frequency_score	0.0293
amount_stddev_90d	0.0289
risk_txn_volatility_90d	0.0285
debit_sum_30d	0.0267
eng_login_count_7d	0.0265
temp_weekday_txn_ratio_90d	0.0259

Hyperparameters

Parameter	Value
max_depth	4
learning_rate	0.05
n_estimators	50
min_child_weight	2
reg_lambda	1.0
subsample	0.8
colsample_bytree	0.8
scale_pos_weight	18.6946

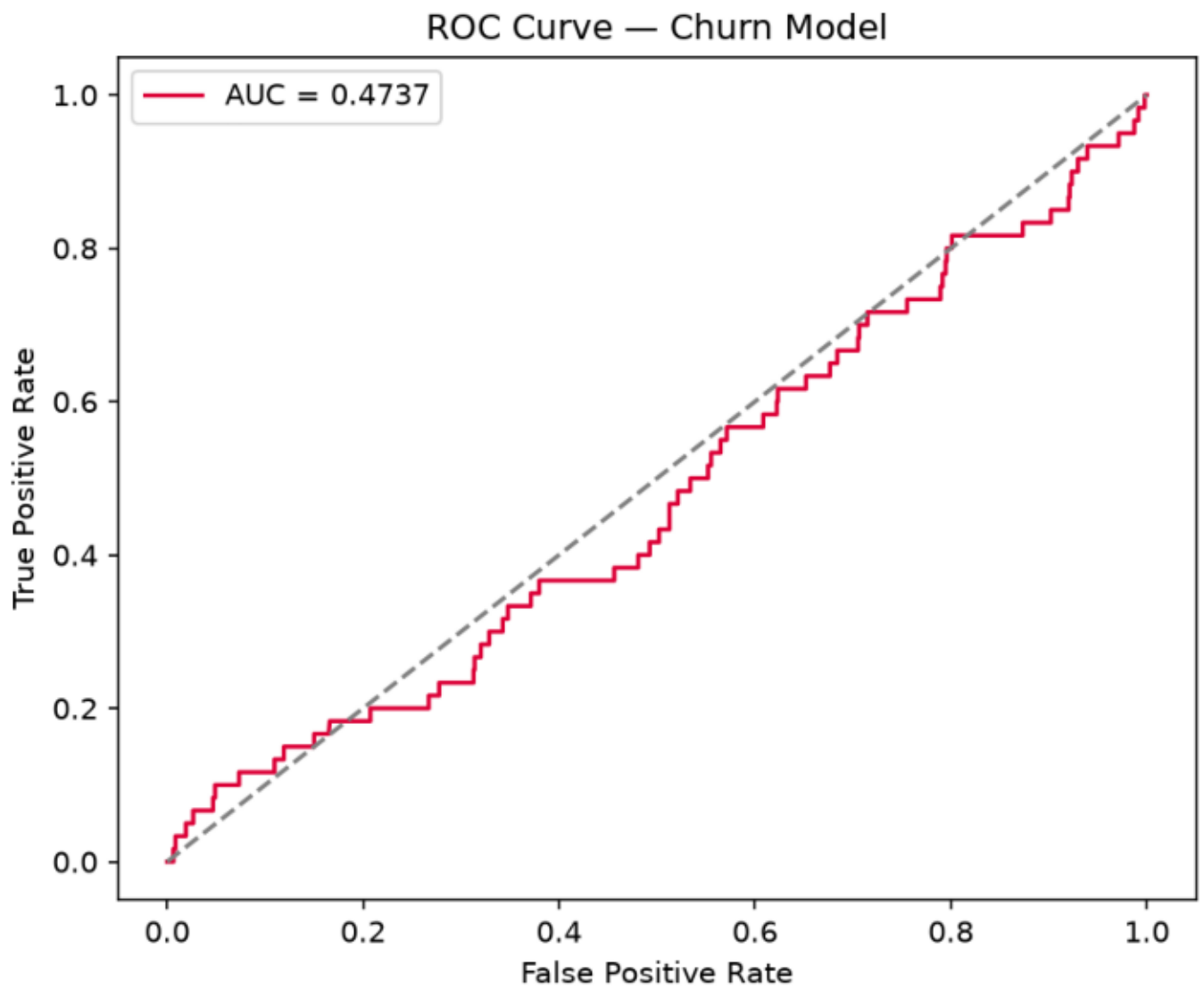
Diagnostics & Drift

train AUC : 0.8914
test AUC : 0.4737
overfit gap : 0.4177 (flag=True)
test missing ratio : 0.0001
overconfidence : 0.0

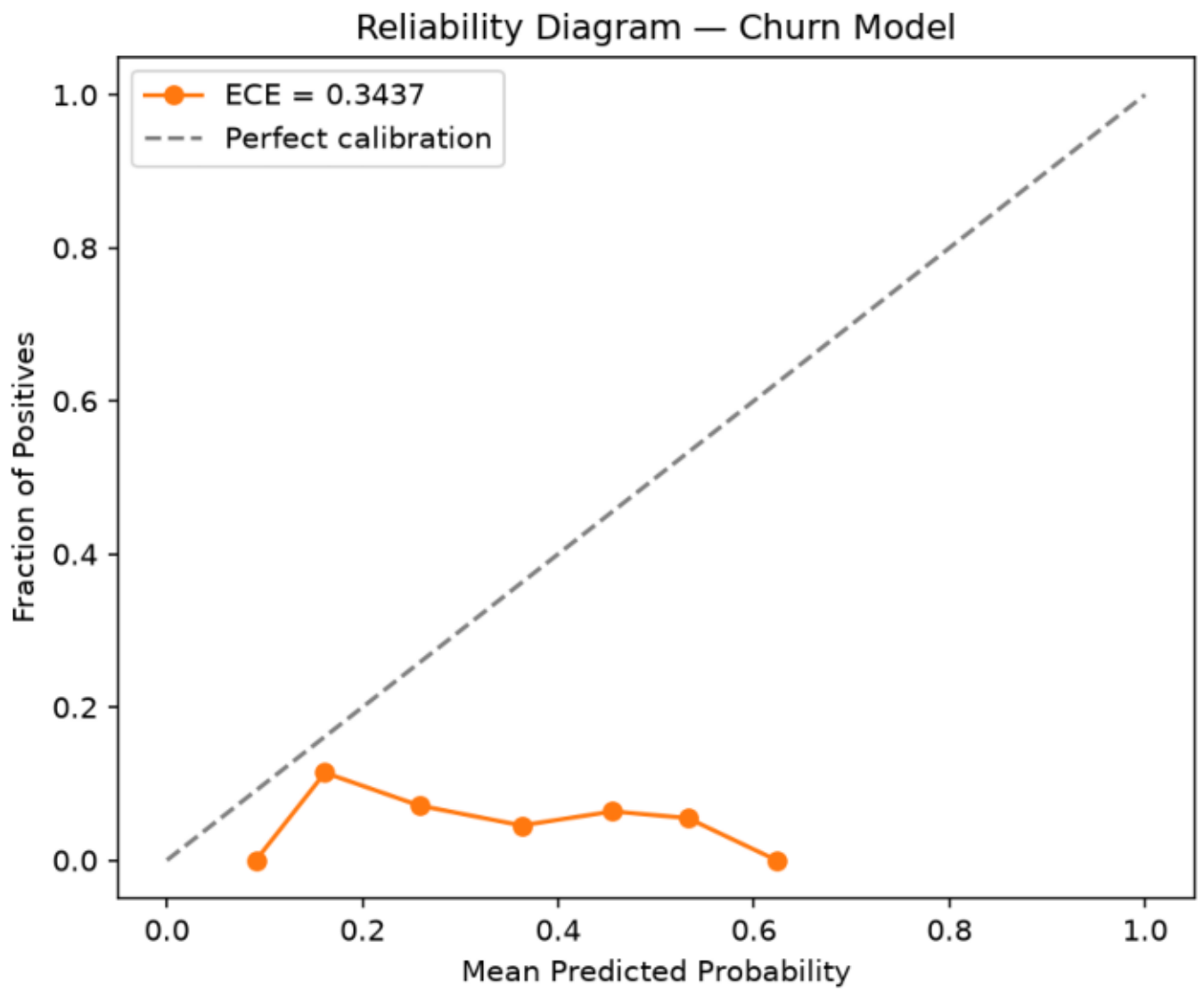
Feature Drift — PSI (max 0.3515, high)

Feature	PSI
total_amount_180d	0.3515
total_amount_90d	0.1606
txn_count_180d	0.0955
days_since_first_txn	0.0935
avg_days_between_txn	0.0703
txn_count_90d	0.0337
temp_weekend_txn_ratio_90d	0.0277
behav_diversity_score	0.0172
fin_median_txn_amount_90d	0.0172
behav_active_days_90d	0.0169

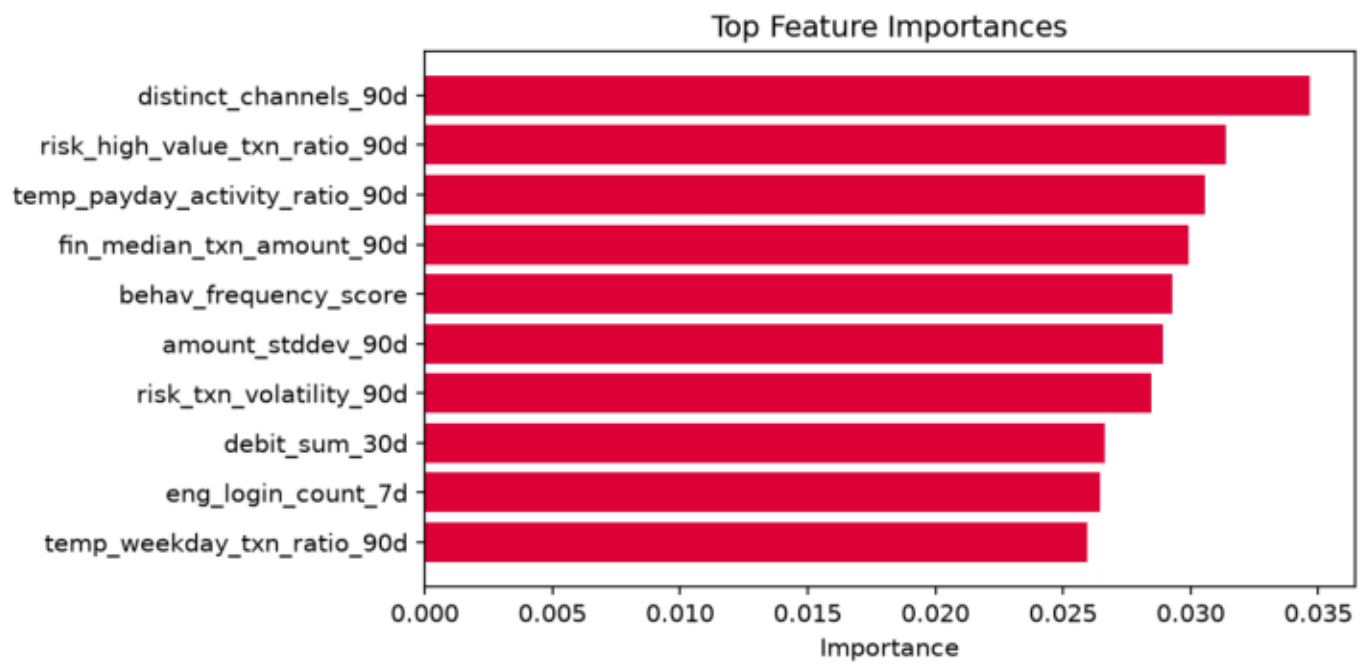
Roc Curve



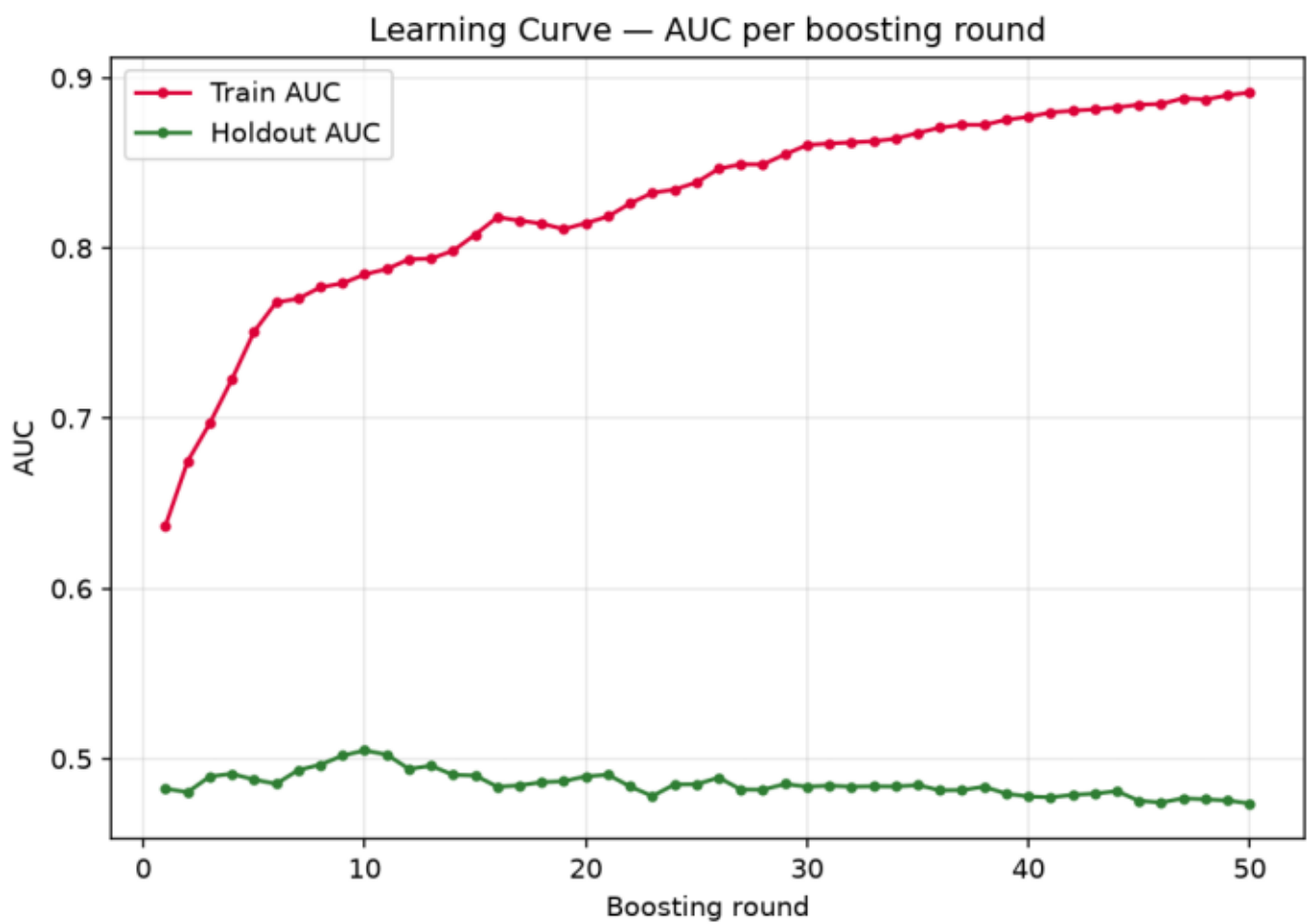
Calibration



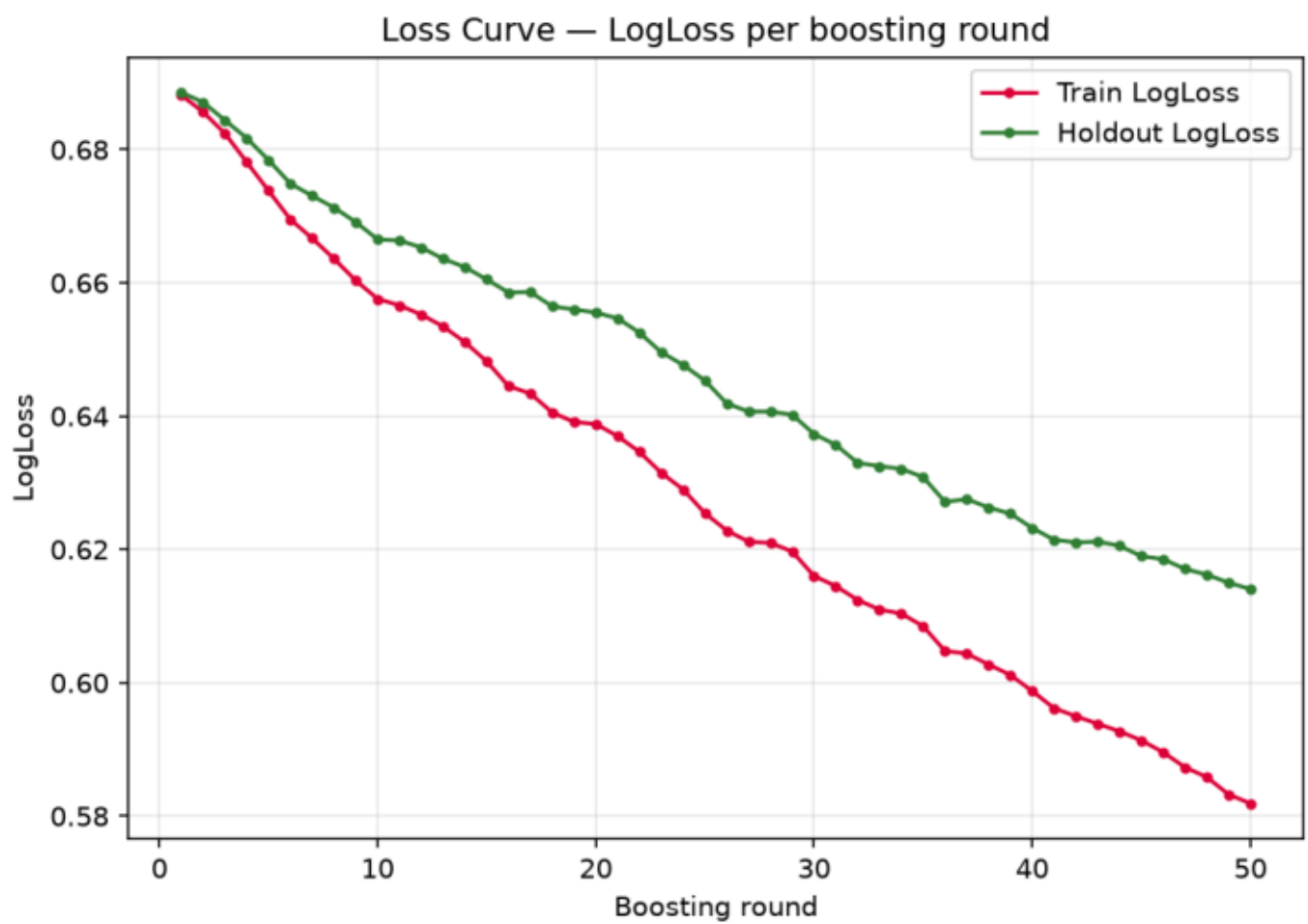
Feature Importance



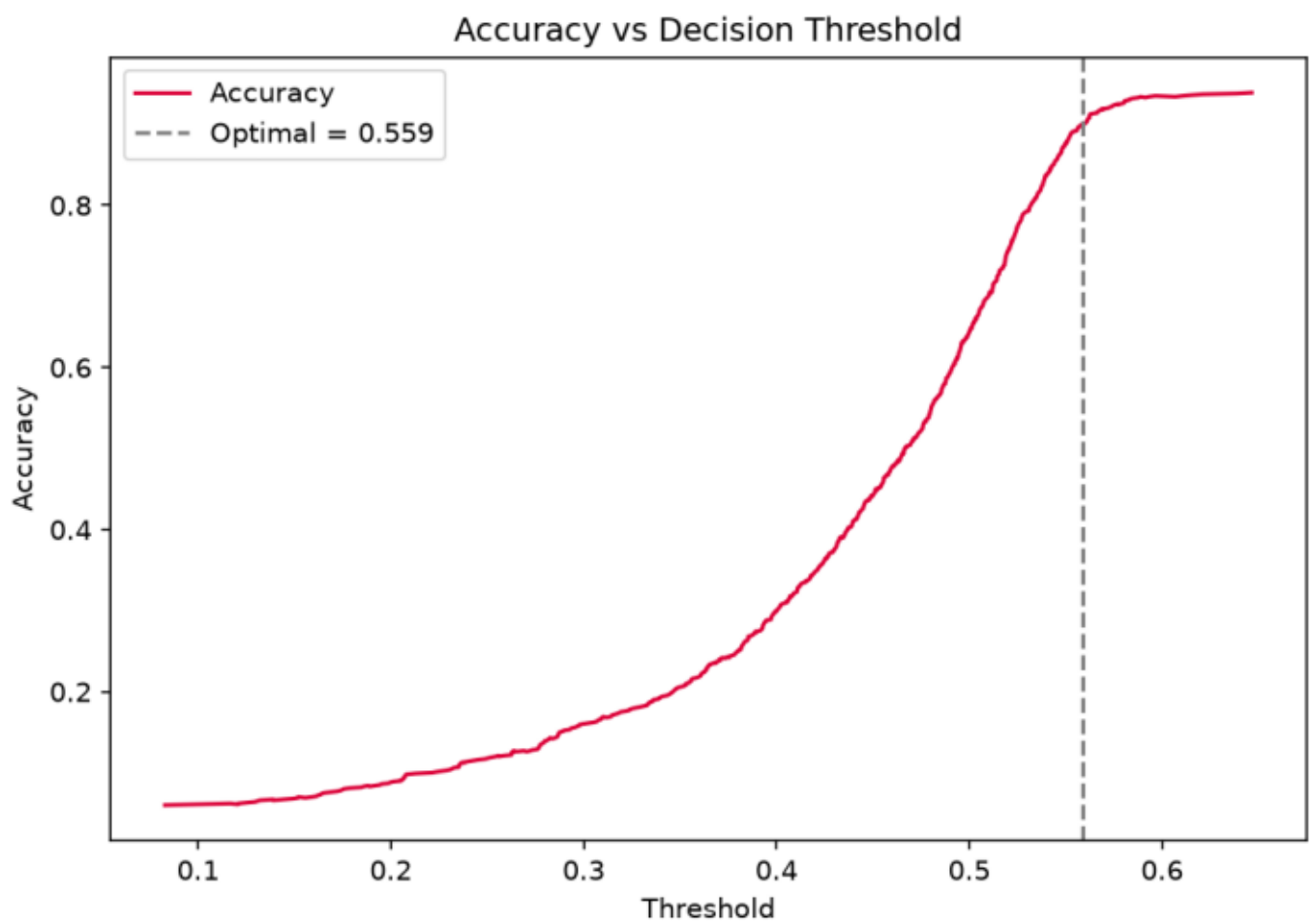
Learning Curve



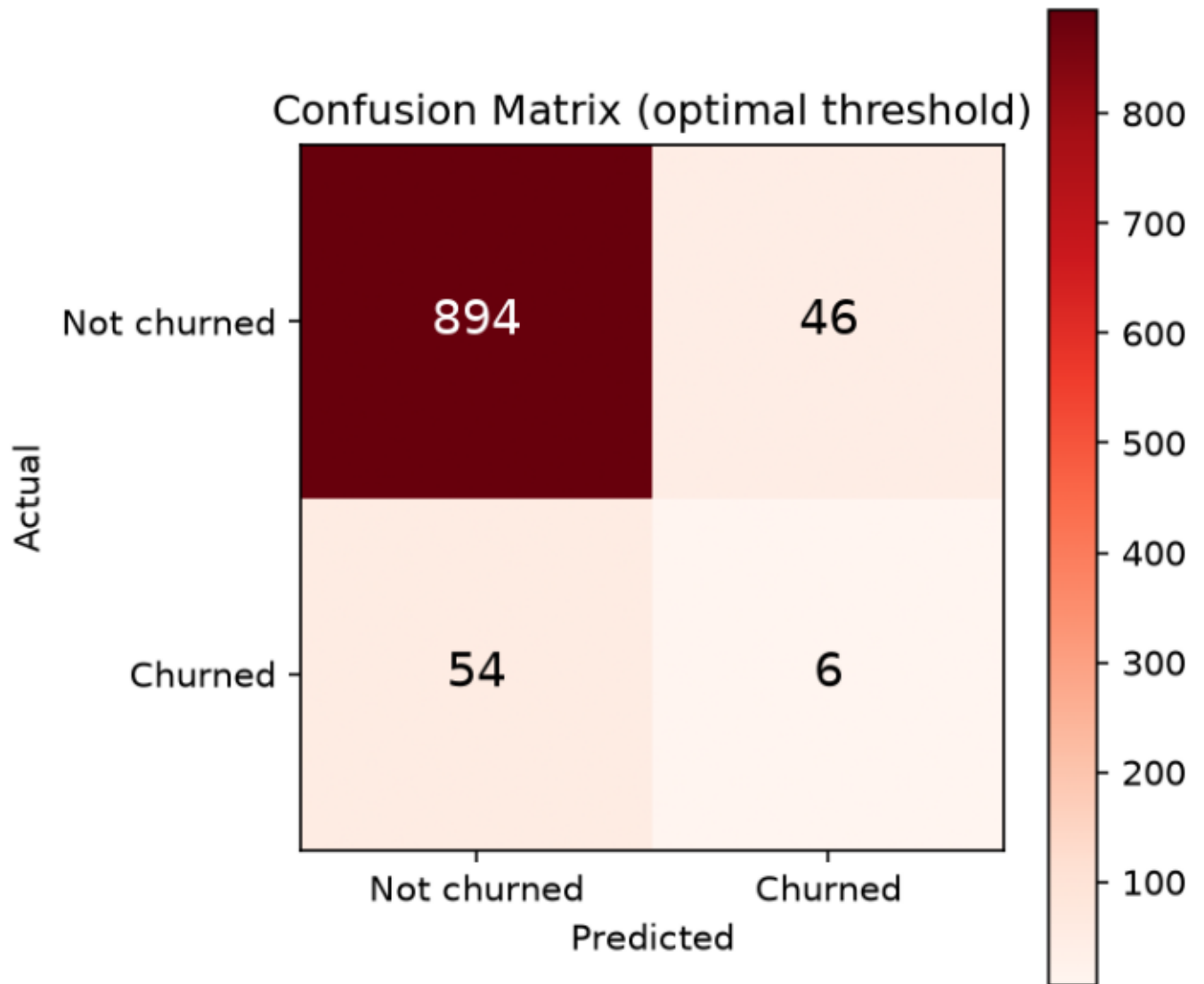
Loss Curve



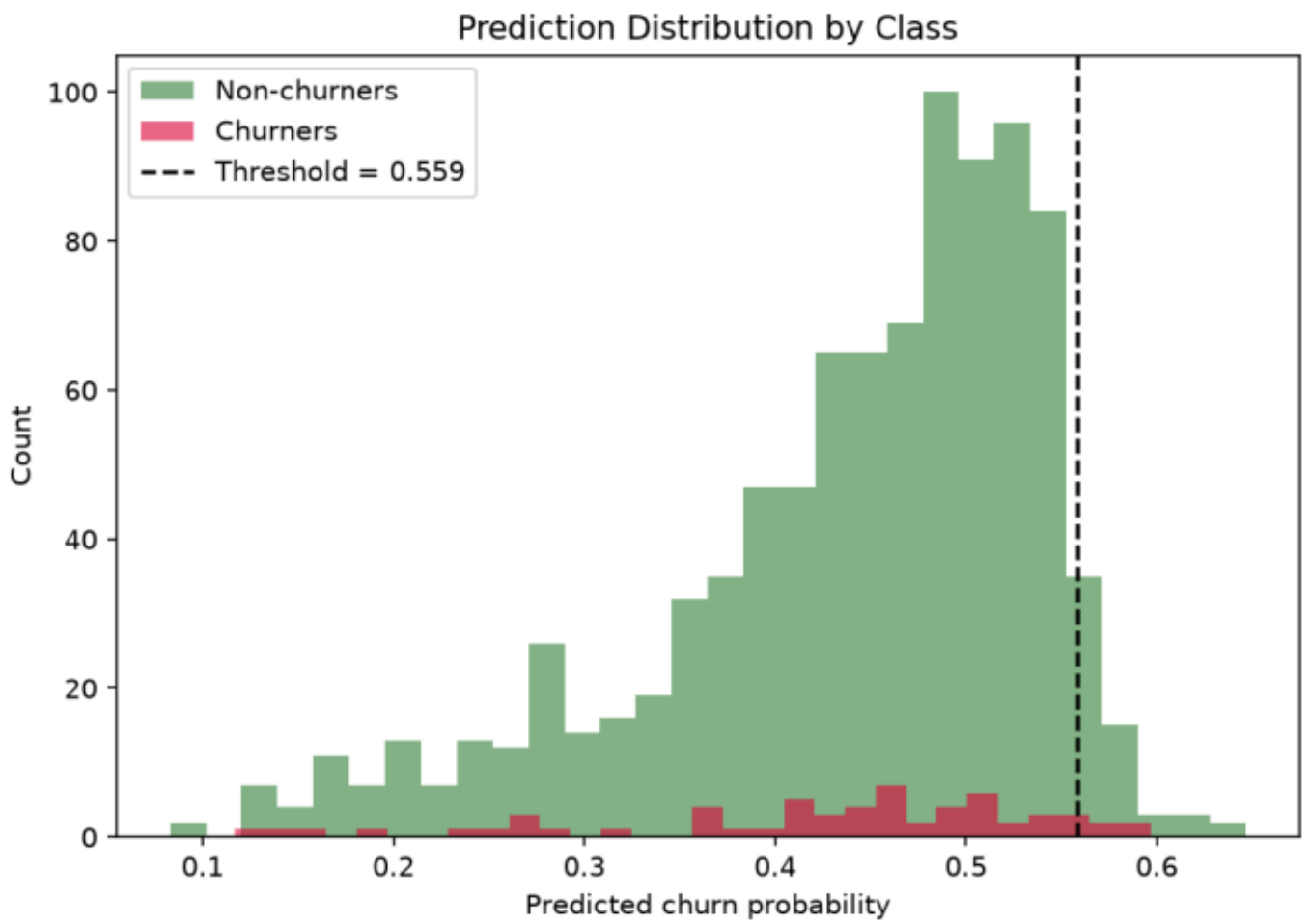
Accuracy Threshold



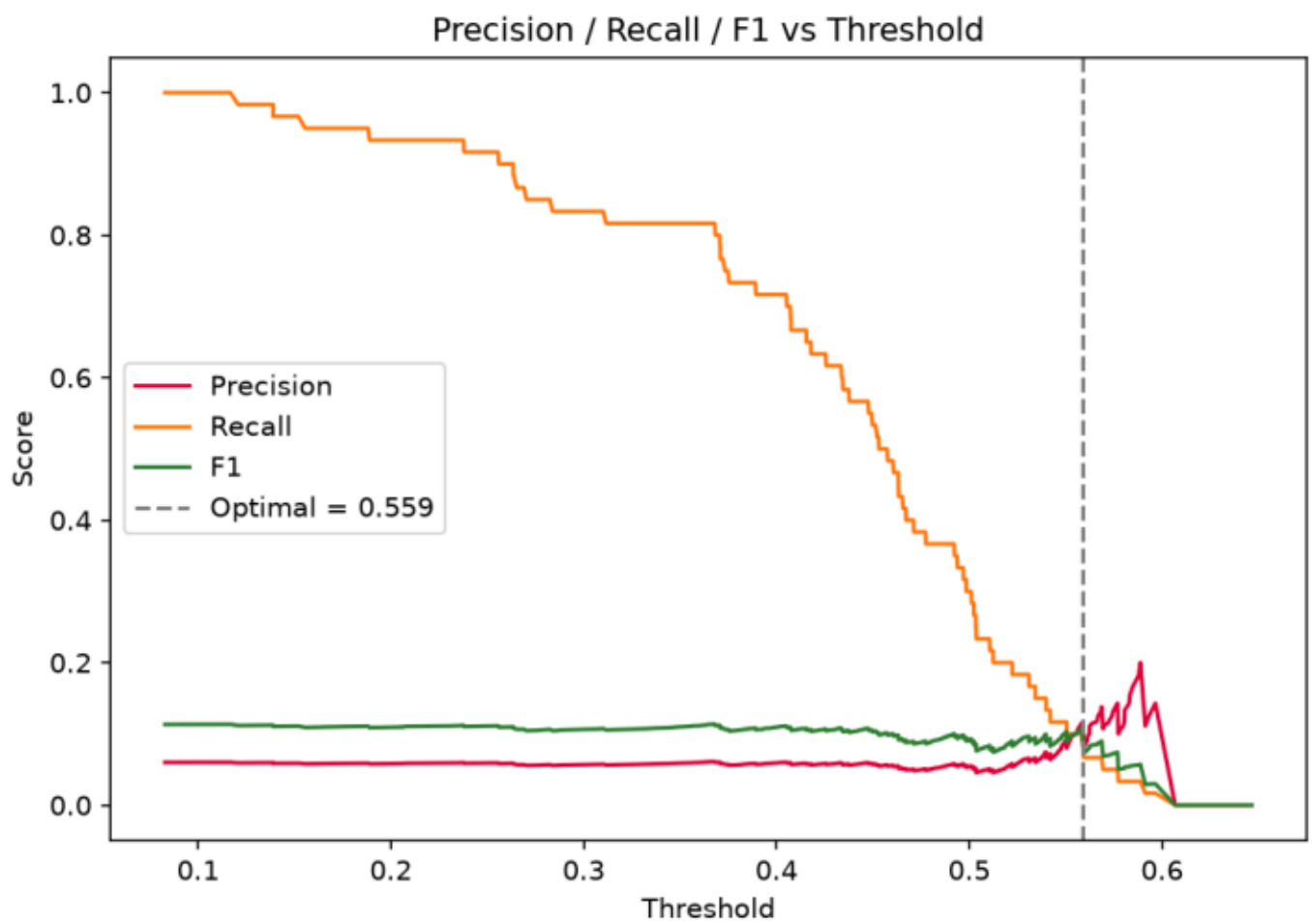
Confusion Matrix



Prediction Distribution



Threshold Analysis



Calibration Comparison

